OLIST E-COMMERCE LOGISTICS ANALYSIS

Business Findings Summary | Brazilian E-Commerce Public Dataset by Olist

Project objective

Analyze order, delivery, seller, customer, regional and product data to identify patterns that may help explain late deliveries and operational performance. The analysis was completed in MySQL 8.0 using 20 SQL case-study queries.

Five key findings

1. Late deliveries vary significantly by destination state	Alagoas (AL) recorded a late-delivery rate of approximately **23.93%**, compared with the overall rate of **8.11%** among delivered orders with valid delivery dates. This identifies AL as a priority for further investigation.
2. Average delivery time differs across states	Roraima (RR) had the longest average delivery time at approximately **29.34 days**, while São Paulo (SP) averaged approximately **8.70 days**. Distance, network design and local delivery conditions may be factors to investigate; the dataset alone does not establish causation.
3. Beauty and health led product revenue	The *beleza_saude* category generated approximately **R$1.26 million** in product revenue, the highest among the categories in the analysis. Revenue here is based on item price and excludes freight.
4. A group of high-volume sellers fell below the 90% on-time threshold	Ten sellers with at least 500 delivered orders had on-time delivery rates below 90%. The lowest observed rate was approximately **87.81%**. These sellers are candidates for deeper operational review.
5. A small set of lanes represented about half of lane-level late counts	The four highest-volume seller-origin/destination-state combinations accounted for approximately **51%** of the lane-level late-order counts. These lanes may be useful priorities for targeted analysis.

Recommended business follow-ups

Prioritize a root-cause review of late deliveries to AL and other high-delay states; review high-volume sellers below the 90% on-time threshold; and investigate the highest-impact origin/destination combinations. Validate these patterns with more detailed operational data before making policy or carrier decisions.

Scope and data limitations

The Olist dataset covers historical Brazilian e-commerce activity (2016–2018). It does not include carrier identifiers or a predefined region field. Seller-level delivery performance is therefore used only as a proxy for carrier performance, and customer state is used for geographic analysis. A single order may contain items from multiple sellers; consequently, lane-level late counts can count the same order in more than one seller/state lane. Freight value is used as a shipping-cost proxy. Findings are descriptive, not causal.

Tools: MySQL 8.0 CE and MySQL Workbench 8.0 CE | **Analysis:** 20 SQL queries including joins, CTEs and window functions